Design note: a global model of where unrest leads to conflict

Question. *Where* does recorded unrest lead to later conflict (escalation), and where do the two only share episodes? The model answers it with escalation coefficients and a shared-episode loading that vary in space, estimated in one model over five regions.

Principles. (1) Effects of observed events go in the mean. (2) The latent part absorbs unobserved structure only. (3) Every parameter has a 95% interval; a component, including any spatial variation, stays only if its interval supports it. (4) No latent coupling between fields.

0. Data, regions and output

Data. Cells s (0.5°), weeks t; $Y_{\mathrm{C}st}, Y_{\mathrm{U}st} \in \{0,1\}$: at least one recorded conflict or unrest event. Covariates $\boldsymbol{x}_{st}$ (climate anomalies, population, national indicators, missingness indicators), the country $c(s)$, histories from the outcomes.
Regions $r(s)$, by country, each from the start of ACLED coverage to 2025: Africa (1997), Middle East (about 2016), South and Southeast Asia (about 2010), Latin America and the Caribbean (2018); Europe (2018–2020) in the Supplementary Material only. Cell-weeks before coverage are not observed and do not enter the likelihood.
Output. Maps of escalation $\kappa_{\mathrm{U}\to\mathrm{C}}(s)$ and $\kappa_{\mathrm{C}\to\mathrm{U}}(s)$ and of the loading $\lambda(s)$, with intervals; the escalation kernel over lags and distances; maps of the latent fields; held-out predictions by region.

1. Data level

Given the linear predictors, $Y_{\mathrm{C}st}$ and $Y_{\mathrm{U}st}$ are independent Bernoulli variables, $\operatorname{logit} P(Y_{jst}=1)=\eta_{jst}$, independent over cells and weeks.

2. Linear predictors

$$\begin{aligned}
\eta_{\mathrm{C}st} &= \boldsymbol{x}_{st}^{\top}\beta_{\mathrm{C}} + \alpha_{\mathrm{C},r(s),y(t)} + \delta_{\mathrm{C}}\, h_{\mathrm{C}st} + \delta_{\mathrm{C}}^{nb}\,\bar{h}_{\mathrm{C}st} + \textstyle\sum_{b,\ell}\kappa_{\mathrm{U}\to\mathrm{C}}(b,\ell)\,H_{\mathrm{U}st}(b,\ell) + \gamma_{\mathrm{U}\to\mathrm{C}}(s)\,H_{\mathrm{U}st}(1,0)\\
&\quad + u_{\mathrm{C},c(s)} + e_{\mathrm{C}s} + w_{\mathrm{C}s} + v_{\mathrm{C}gk},\\
\eta_{\mathrm{U}st} &= \boldsymbol{x}_{st}^{\top}\beta_{\mathrm{U}} + \alpha_{\mathrm{U},r(s),y(t)} + \delta_{\mathrm{U}}\, h_{\mathrm{U}st} + \delta_{\mathrm{U}}^{nb}\,\bar{h}_{\mathrm{U}st} + \textstyle\sum_{b,\ell}\kappa_{\mathrm{C}\to\mathrm{U}}(b,\ell)\,H_{\mathrm{C}st}(b,\ell) + \gamma_{\mathrm{C}\to\mathrm{U}}(s)\,H_{\mathrm{C}st}(1,0)\\
&\quad + u_{\mathrm{U},c(s)} + e_{\mathrm{U}s} + w_{\mathrm{U}s} + \lambda(s)\,v_{\mathrm{C}gk} + v_{\mathrm{U}gk},
\end{aligned}$$

with coarse cell $g=g(s)$ and time step $k=k(t)$.

- **Year effects by region**, $\alpha_{j,r,y}$: absorb changes in ACLED's reporting, which differ between regions.
- **Own histories** (as now): $h_{jst}=\log(1+\text{events of type } j \text{ in cell } s \text{ in weeks } t-4,\dots,t-1)$ and $\bar{h}_{jst}$, its mean over the rook neighbours.
- **Escalation kernel**: $H_{\mathrm{U}st}(b,\ell)$ is the mean over the cells of distance ring ℓ (R_0: the cell; R_1: rook neighbours; R_2: 50–300 km) of $\log(1+\text{unrest events in lag band } b)$, with B_1 = 1–4, B_2 = 5–13 and B_3 = 14–26 weeks. Its 9 coefficients per direction are common to all regions (the shape of escalation).
- **Spatially varying escalation**: the short-run, same-cell effect is $\kappa_{\mathrm{U}\to\mathrm{C}}(s)=\kappa_{\mathrm{U}\to\mathrm{C}}(1,0)+\gamma_{\mathrm{U}\to\mathrm{C}}(s)$, and likewise $\kappa_{\mathrm{C}\to\mathrm{U}}(s)$. Only this term varies, because it carries most of the data.

3. Latent level

Static random effects, one independent set per outcome j: country effects $u_{j,c}\sim\mathcal{N}(0,\sigma_{u,j}^2)$; cell effects $e_{js}\sim\mathcal{N}(0,\sigma_{e,j}^2)$; a Matérn field w_j with sea barriers.

Space-time fields, independent: v_{C} (conflict) and v_{U} (unrest only), each

$$v_{jgk}=\varphi_j\,v_{jg,k-1}+\sqrt{1-\varphi_j^2}\;\varepsilon_{jgk},\qquad (\varepsilon_{j1k},\dots,\varepsilon_{jGk})\sim\text{Matérn}(\rho_j,\sigma_j)\text{ with sea barriers},$$

independent over steps k. On a sphere-wide lattice the fields of different continents are independent through the sea barriers, but a field runs across land borders (Africa–Middle East–Asia).

Spatial variation, two versions. Each is a modelling choice to be settled by the prototype and the simulation (Section 6).

	piecewise by region	smooth
escalation $\gamma(s)$	γ_r: fixed effects, one per region	Matérn field (spatially varying coefficient)
loading $\lambda(s)$	λ_r: one hyperparameter per region	$\sum_{m=1}^{M} \theta_m b_m(s)$, $M \approx 5$–10 smooth basis functions

Both keep the model latent Gaussian. $\gamma(s)$ enters the mean linearly. For $\lambda(s)$, the loading sits in the prior precision of the pair $(v_C, \lambda v_C)$, as in the current model, and the log-determinant does not depend on it, whether λ is a region value or a basis expansion. A full Gaussian field *in* the loading would break this, so we do not use one.

4. Parameters and priors

Fixed effects ($\mathcal{N}(0, 10^4)$): β_j; $\alpha_{j,r,y}$; δ_j, δ_j^{nb}; 2×9 kernel coefficients; γ_r in the piecewise version.
Hyperparameters: static, 8 (as now); space-time fields, 6 ($\rho_j, \sigma_j, \varphi_j$); loadings, 4 regions ($\lambda_r$) or M basis coefficients, $\mathcal{N}(0, 2^2)$; in the smooth version, the SD and range of each γ field, 4, with PC prior $P(\sigma_\gamma > 0.2) = 0.05$, which shrinks towards no variation. In total 18 (piecewise) to about 28 (smooth).

5. What each component answers, and the evidence it needs

- $\kappa_{U \to C}(b, \ell)$: how escalation decays with lag and distance. Each coefficient with its interval.
- $\gamma_{U \to C}$: where escalation is stronger or weaker. Evidence of variation: the interval for σ_γ well away from zero (smooth), or intervals for regional differences excluding zero (piecewise). Otherwise a constant κ.
- $\lambda(s)$: where conflict and unrest share episodes. Same rule. If every λ_r interval covers zero, the two outcomes are fitted separately.
- v_U, w_j, u_j, e_j: as now, each with its interval (all supported in Africa and the Middle East).

6. Validation before the global fit

1. **Prototype on Africa and the Middle East** (existing data, three regions: Africa, Middle East, Afghanistan and Pakistan), warm-started from the current extended model: region loadings; region loadings with a smooth $\gamma(s)$; region loadings with regional γ_r. Intervals for every hyperparameter. With one region, it reproduces the current model exactly.
2. **Simulation** (S2 design, two regions, 20 replicates): (i) constant κ and λ, where no spurious variation should be found; (ii) $\kappa(s)$ varying (SD 0.3) and $\lambda_r = 0.3$ and 1.2, which should be recovered with correct coverage.

7. Computation

About 60 million cell-weeks per outcome. Approximations, each checked against a full fit on a region where one is affordable and kept only if the conclusions are unchanged: case-control subsampling of empty cell-weeks; space-time fields on 2° cells and 13–26 week steps; intervals from the Hessian at the estimate. Engineering: CHOLMOD on the CRC, and analytic gradients for the outer optimisation (the number of hyperparameters doubles).